

CHAPTER 5

AUXILIARY VIEW

5.1: *Introduction*

In orthographic projection sometimes it happens that some surfaces of the machine parts are not parallel to the plane of projection i.e. they are at an angle. These surfaces are called inclined surfaces. The regular orthographic view of the inclined surface does not give the exact shape; it gives the distorted shape of the surface. Thus the clear feature of the inclined surface of the object cannot be obtained from the regular view. As a result the main objective of the drawing is lost. In order to overcome this problem an auxiliary view is necessary.

A picture plane is assumed parallel to the inclined surface called auxiliary projection plane and orthographic projection on that plane is made. It gives the true shape of the inclined surface and preserves the objective of the drawing. Sometimes an auxiliary view replaces one of the regular views. For some machine part it may happen that some portion of the surface is inclined and the rest portion is not. In that case to represent the total surface of the object partial views are drawn. The partial view is drawn in order to avoid distortion and complication. In auxiliary view, hidden lines are avoided unless they become necessary to represent the view clearly.

5.2: *Generating Auxiliary View*

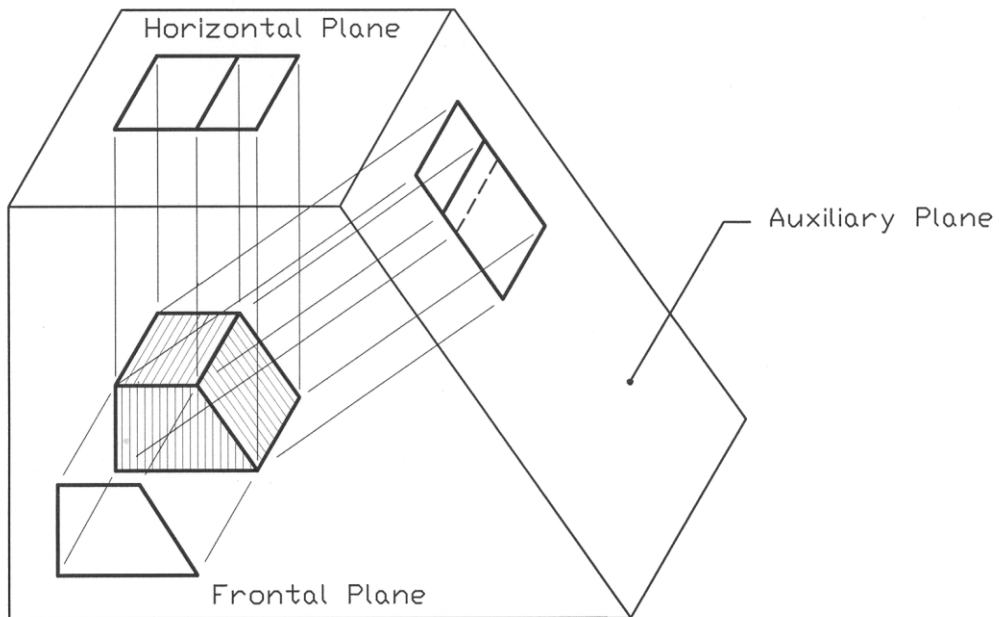


Figure 5.1: Auxiliary View on Inclined Plane

It is described in this section how to generate an auxiliary view. A simple object is placed inside a transparent box (Figure 5.1) each side of which is called the picture plane. The inclined plane which, is parallel to the inclined surface of the object, is called the auxiliary plane as shown in the figure. Other planes are horizontal and frontal planes. In Figure 5.2 the views of the object with the box in the unfolding condition have been shown while in Figure 5.3 the relative positions of the views have been provided.

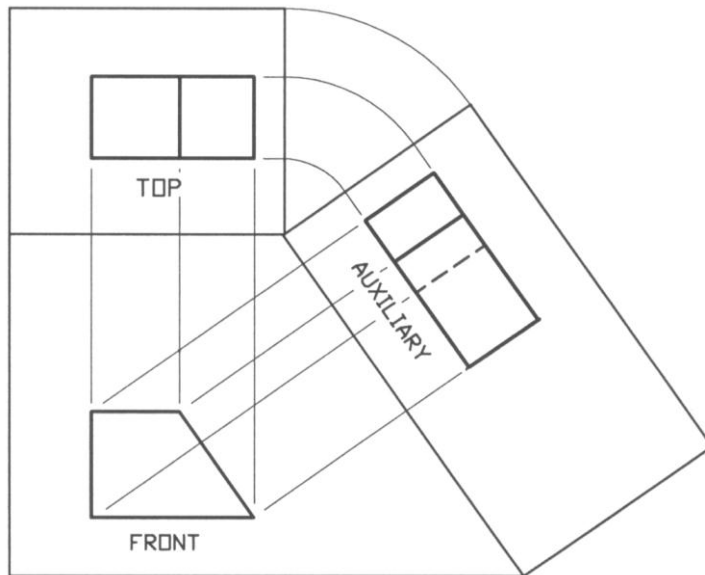


Figure 5.2: Views with Box in Unfolding Condition

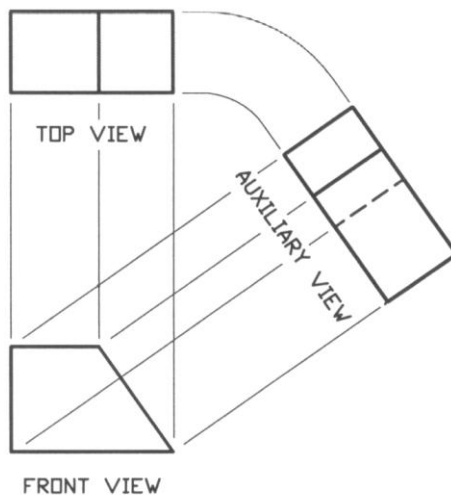


Figure 5.3: Relative Positions of Views

In order to give further example, another object with inclined surface having some complication has been chosen which can be seen in Figure 5.4. The regular orthographic views of the object with inclined surface (Figure 5.4) are given in Figure 5.5. While the necessary views of the same object including an auxiliary view have been presented in Figure 5.6.

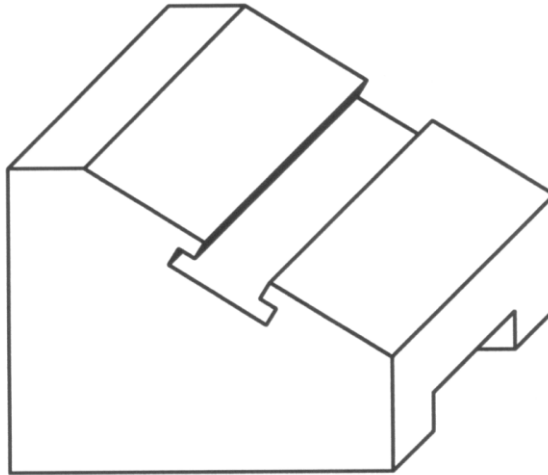
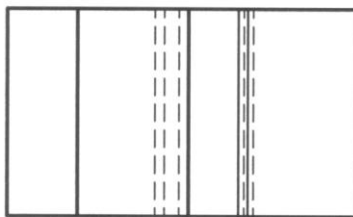
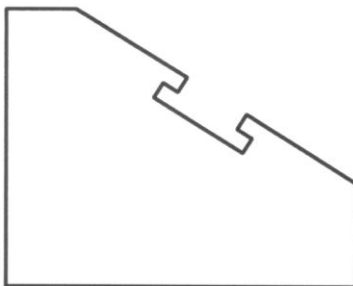


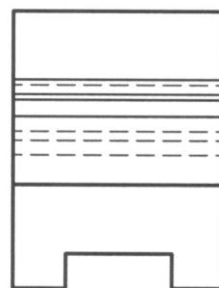
Figure 5.4: Object with Inclined Surface



TOP VIEW



FRONT VIEW



RIGHT SIDE VIEW

Figure 5.5: Regular Orthographic Views

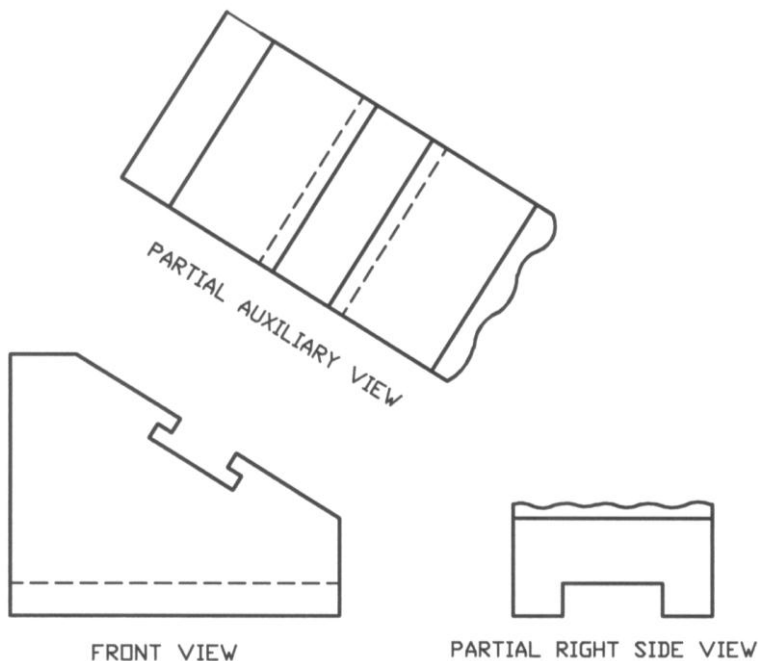


Figure 5.6: Auxiliary View

It is obvious from Figure 5.5 that the regular orthographic views cannot produce the true shape of the object for the inclined surface. As such it becomes very difficult to use them as the detail drawing to manufacture. Because they do not give the clear picture about the inclined surface and in addition the size of the inclined surface is shortened. In order to overcome this problem, auxiliary view becomes essential as shown in Figure 5.6.

It can be observed from Figure 5.6 that, the top view has been dropped and it has been replaced by auxiliary view. Top view is no more necessary for this particular case. It is also seen from this figure that the right side view has been drawn partially. As such, it is called partial right side view.

The auxiliary view gives the true shape of the inclined surface of the object and it becomes clear for understanding. If the auxiliary view includes the feature of the inclined portion of the object only omitting the view of the non-inclined portion (if there is any), it is called the partial auxiliary view. To eliminate distortion a partial auxiliary view is used. On the other hand, if the entire surface is taken under consideration the view becomes a complete auxiliary view. Sometimes a complete auxiliary view is drawn where no distortion appears. Usually a break line is used to indicate the imaginary break in the partial views.

5.3 Auxiliary View with Circular Feature

In Figure 5.7, an object with inclined surface containing a circular feature is shown. The regular orthographic views of the object (Figure 5.7) are shown in Figure 5.8. While the essential views including an auxiliary view are shown in Figure 5.9.

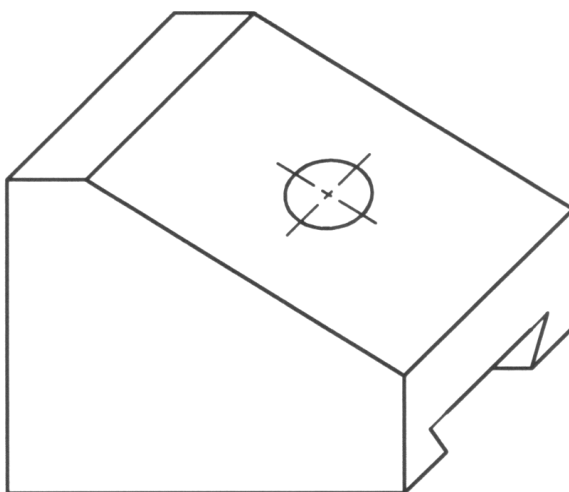
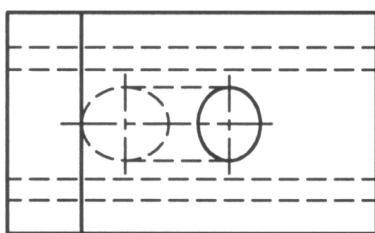
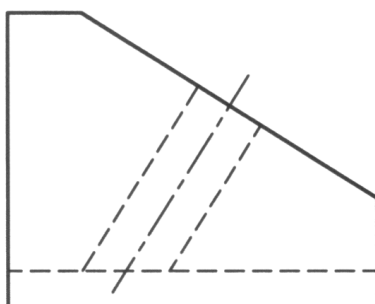


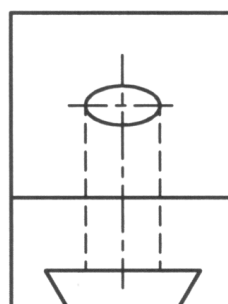
Figure 5.7: Object with Inclined Surface Containing a Hole



TOP VIEW



FRONT VIEW



RIGHT SIDE VIEW

Figure 5.8: Regular Orthographic Views.

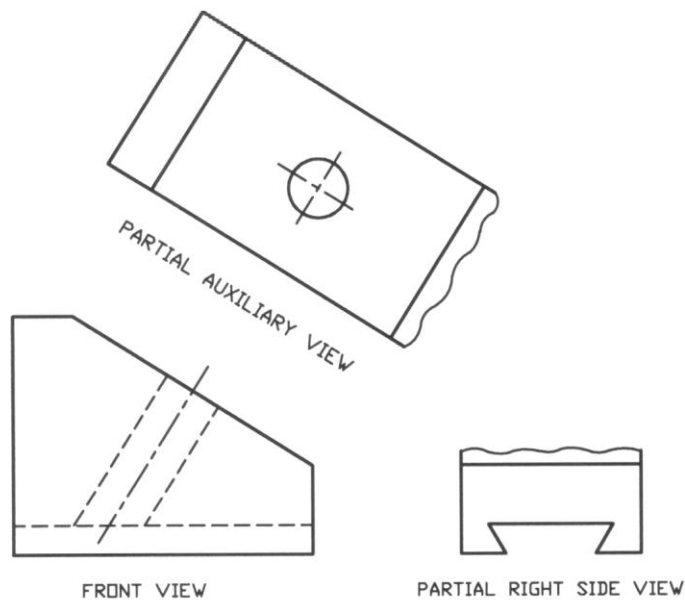


Figure 5.9: Auxiliary View

It is seen from Figure 5.8 that the shape of the hole in the inclined surface has turned into the shape of an ellipse i. e. the true shape of the circle has been distorted. In addition the views have become complicated to understand. To avoid this problem an auxiliary view is required as shown in Figure 5.9. Here a partial auxiliary view and a partial right side view have been drawn. The top view has been omitted because it is not required to represent the object.

5.4: *Multiple Auxiliary Views*

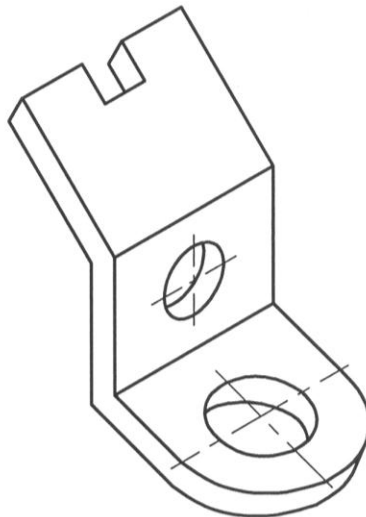


Figure 5.10: Object with two inclined surfaces

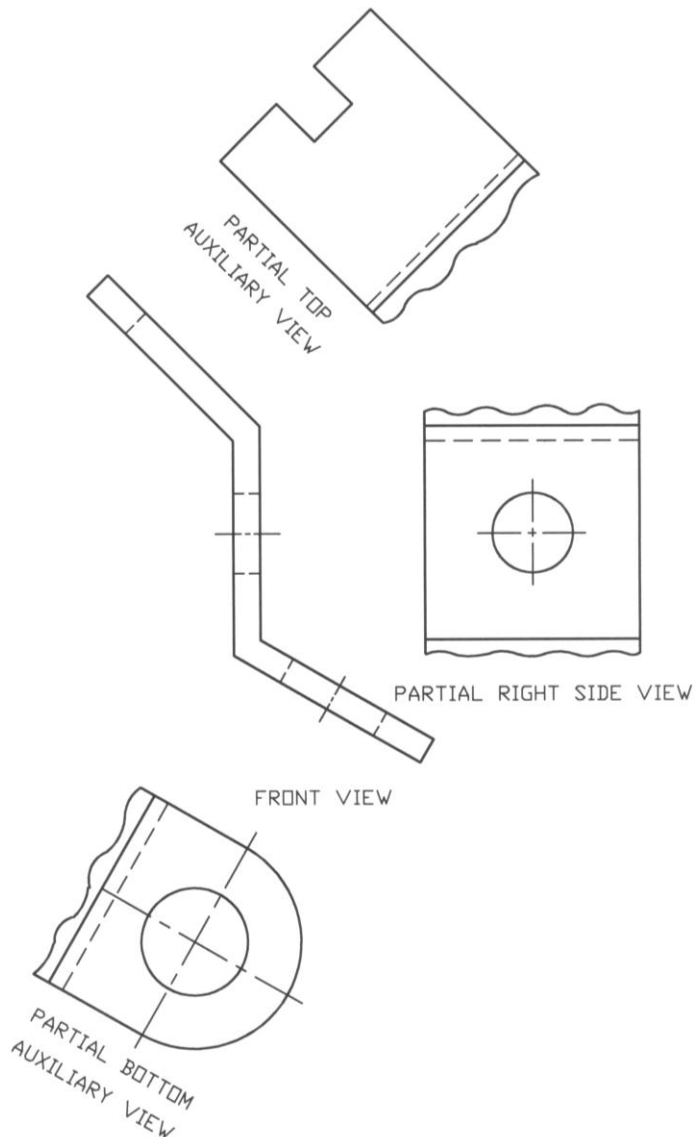


Figure 5.11: Multiple Auxiliary Views

There are some objects, which have several inclined surfaces. They require multiple auxiliary views to represent them clearly. In Figure 5.10 an object with two inclined surfaces are given. In Figure 5.11 the necessary views of the object as shown in Figure 5.10 have been provided. In this figure two auxiliary views have become necessary to represent the two inclined surfaces of the object. Partial auxiliary views have been drawn and a partial right side view has been provided to avoid distortion.

Example Problems

Note: For solutions see the following section of *Solutions for Example Problems*.

Prob. 5.1: Draw front, partial bottom, left and right auxiliary views of the connector plate as shown in Fig. P5.1.

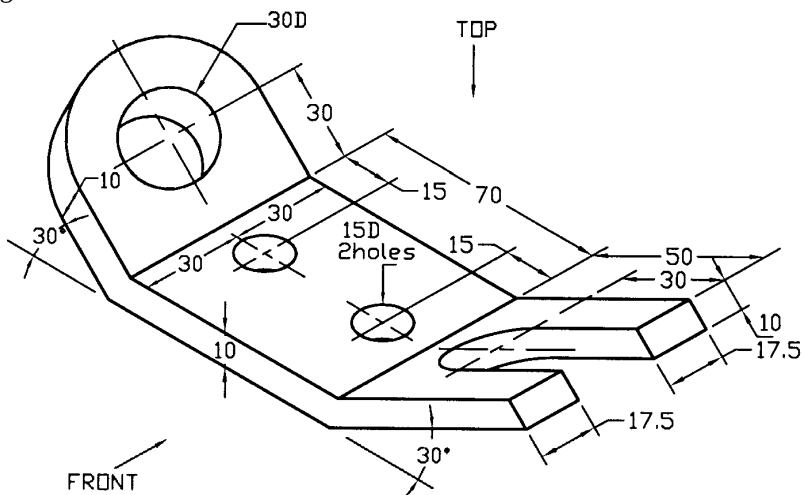


Fig. P5.1

Prob. 5.2: Draw front, partial left, partial top and auxiliary views of the supporting plate as shown in Fig. P5.2.

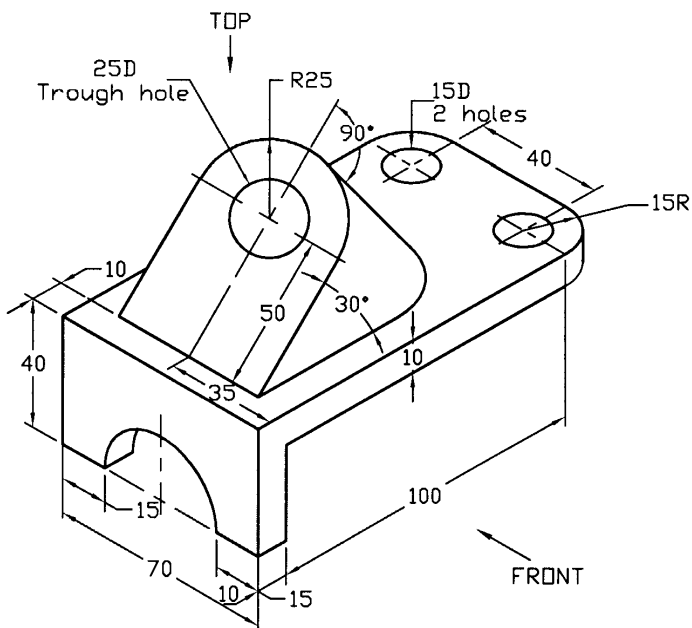


Fig. P5.2

Prob. 5.3: Draw front, partial left side, partial top and auxiliary views of the supporting bracket as shown in Fig. P5.3.

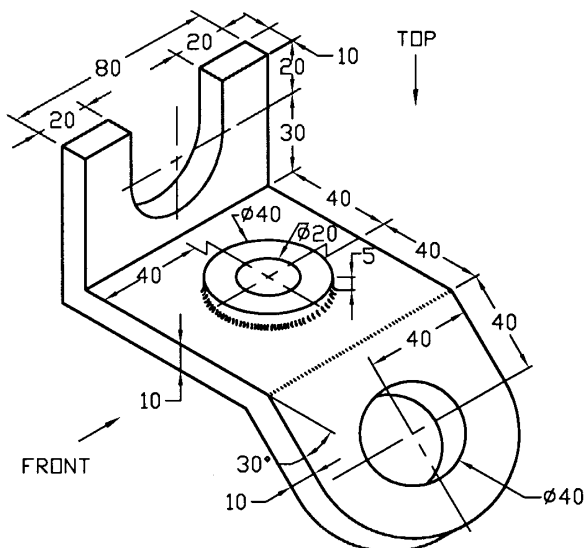


Fig. P5.3

Prob. 5.4: Draw front, partial top and auxiliary views of the holder as shown in Fig. P5.4.

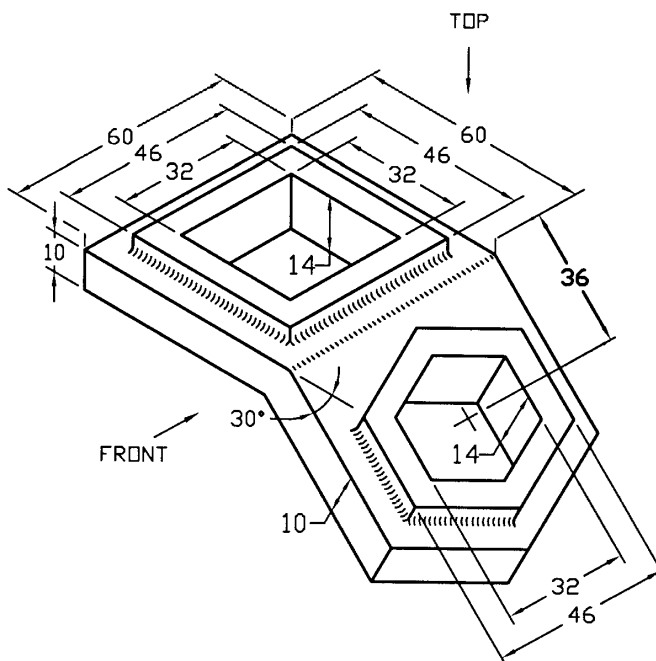


Fig. P5.4

Prob. 5.5: Draw front, partial bottom and auxiliary views of the fixer plate as shown in Fig. P5.5.

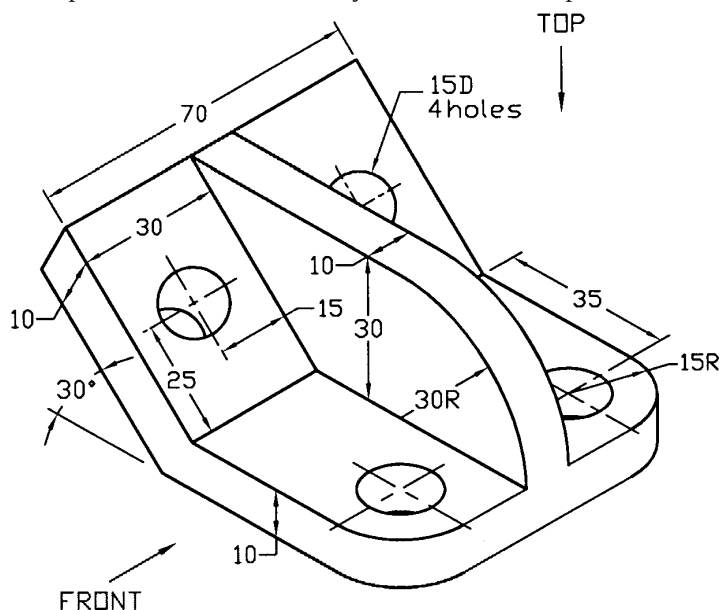


Fig. P5.5

Prob. 5.6: Draw front, partial bottom, partial left side and auxiliary views of the clip as shown in Fig. P5.6.

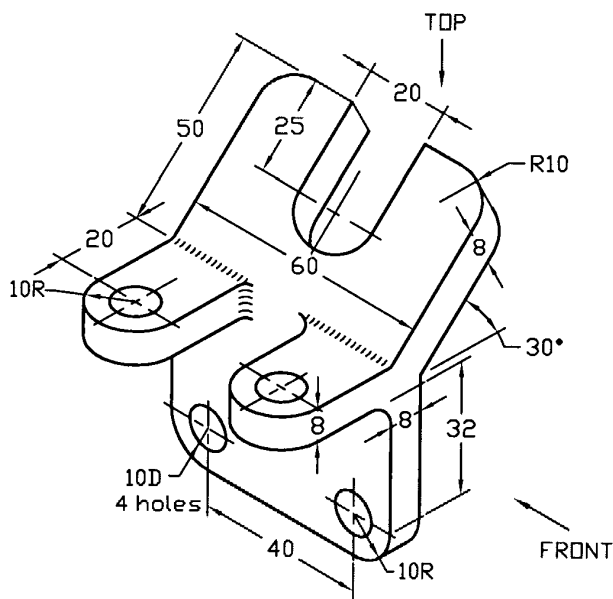


Fig. P5.6

Prob. 5.7: Draw front, partial bottom and auxiliary views of the support bracket as shown in Fig. P5.7.

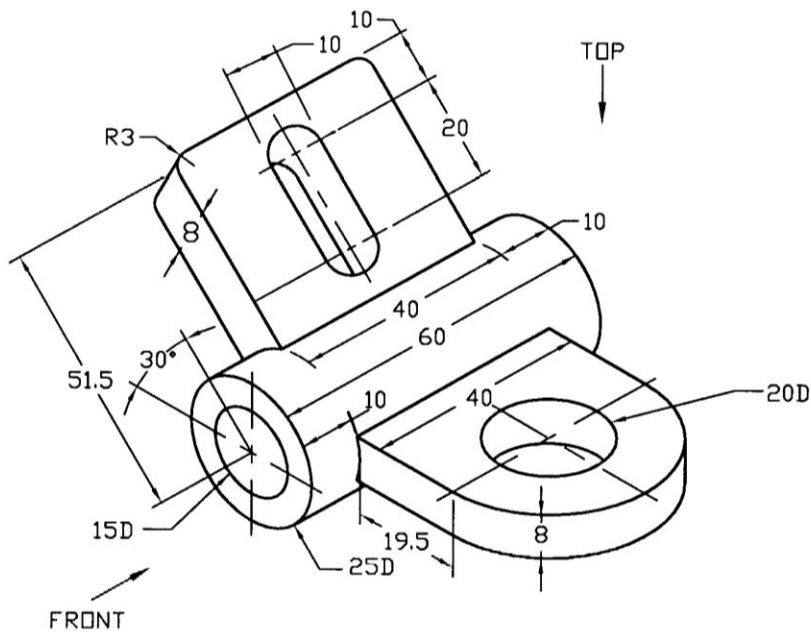


Fig. P5.7

Prob. 5.8: Draw front, partial top and auxiliary views of the angle bracket as shown in Fig. P5.8.

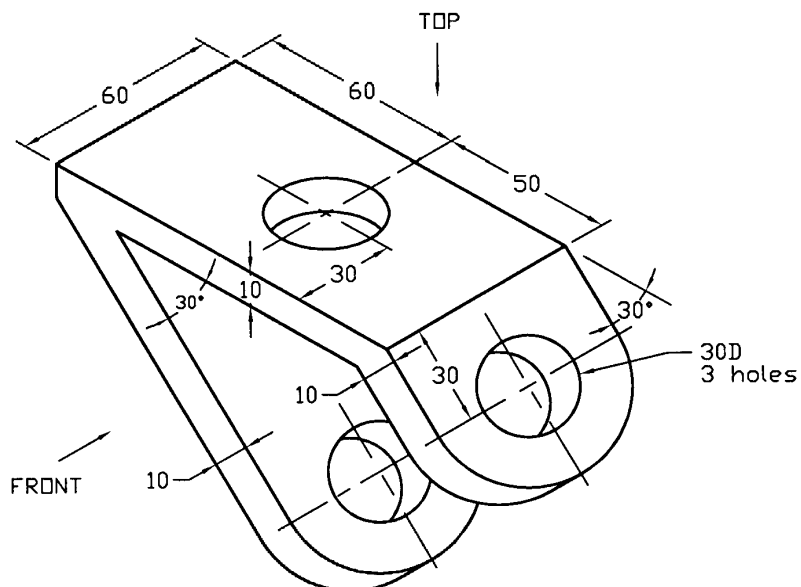


Fig. P5.8

Prob. 5.9: Draw top, partial front and auxiliary views of the wedge block as shown in Fig. P5.9.

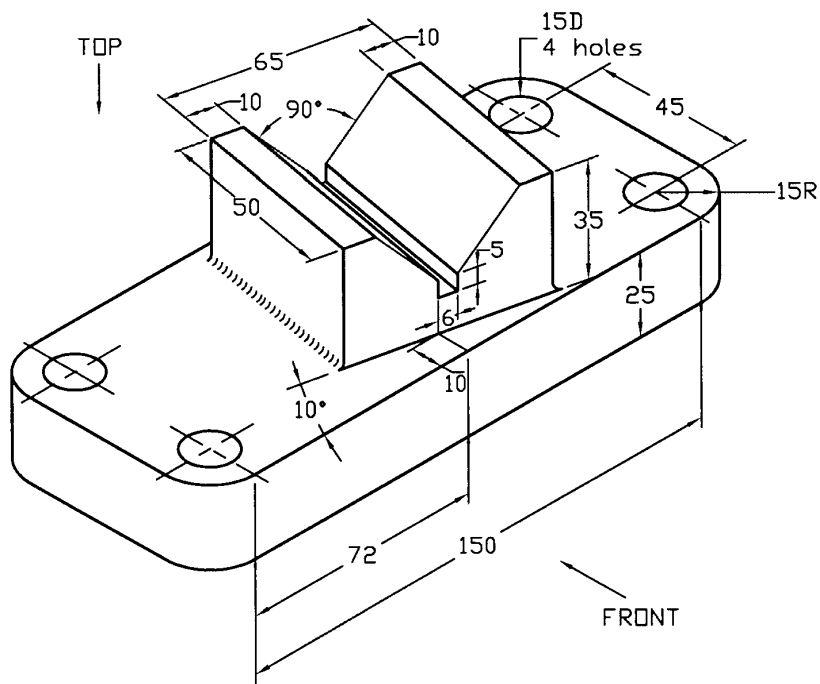


Fig. P5.9

Prob. 5.10: Draw top, partial front and auxiliary views of the connector as shown in Fig. P5.10.

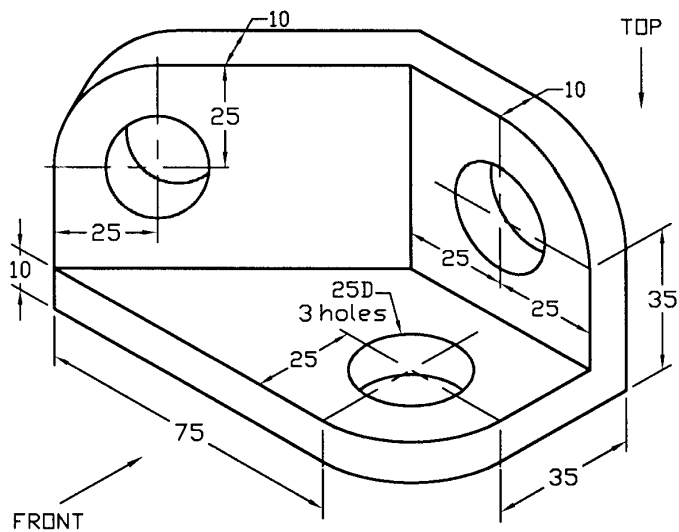


Fig. P5.10

Prob. 5.11: Draw front, partial bottom, left and right auxiliary views of the guide bracket as shown in Fig. P5.11.

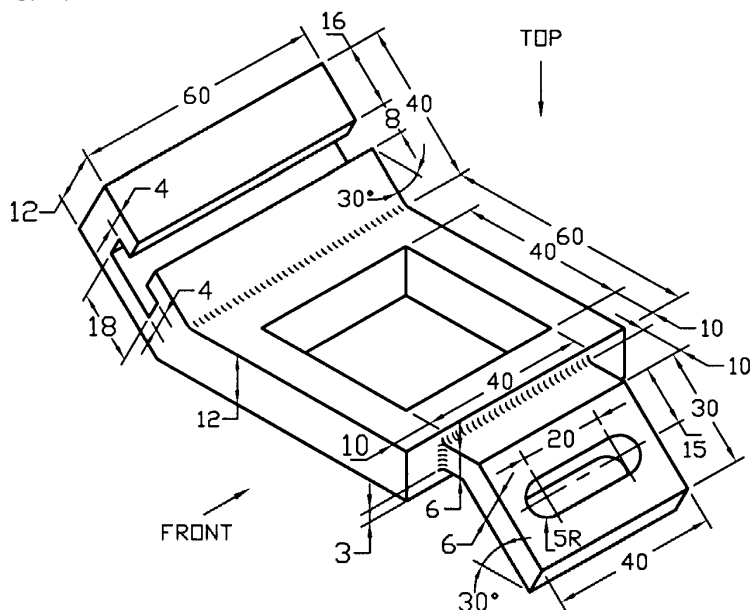


Fig. P5.11

Prob. 5.12: Draw front, partial bottom, partial right side and auxiliary views of the bracket as shown in Fig. P5.12.

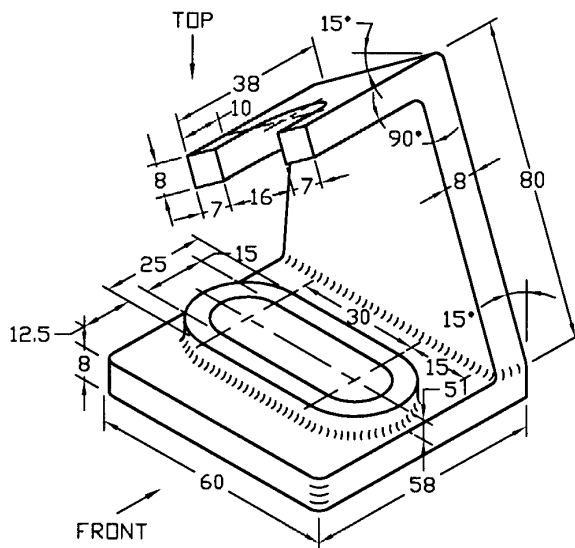
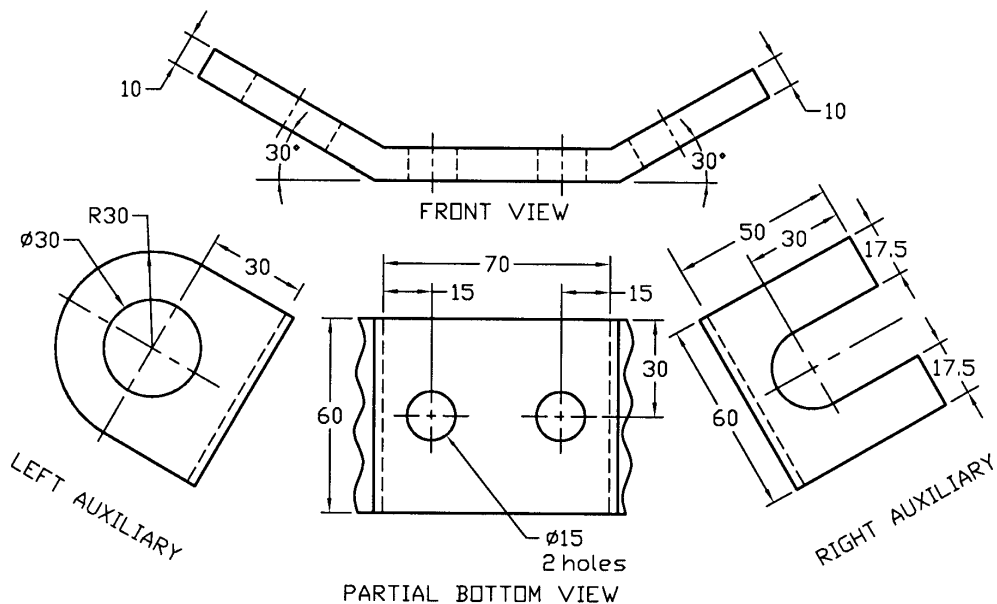
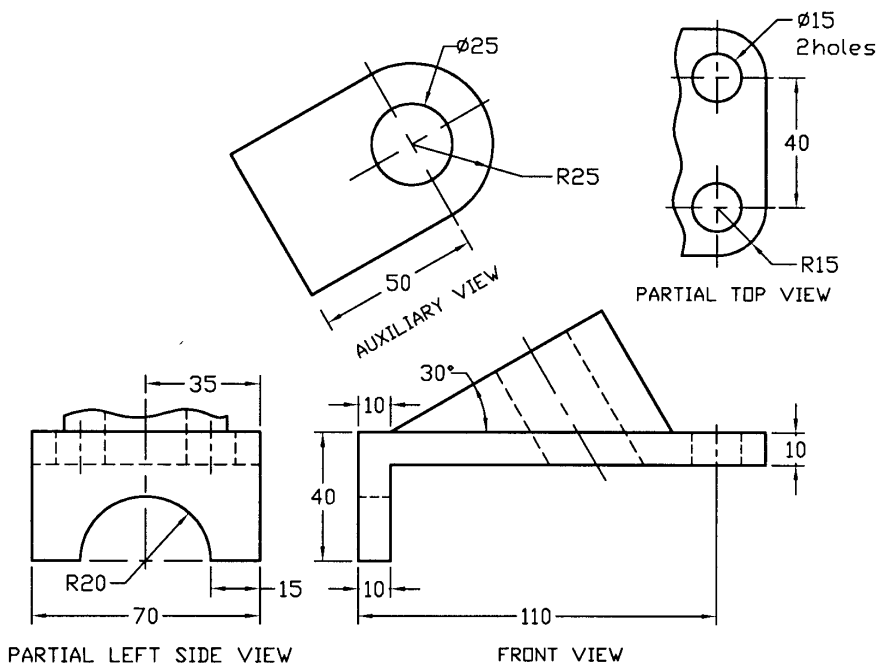


Fig. P5.12

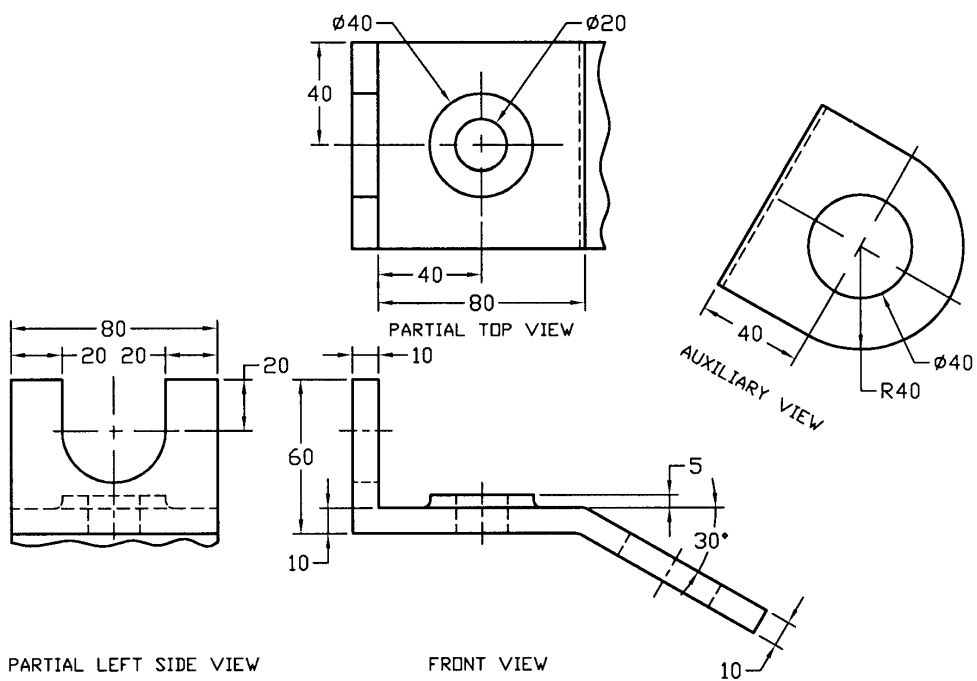
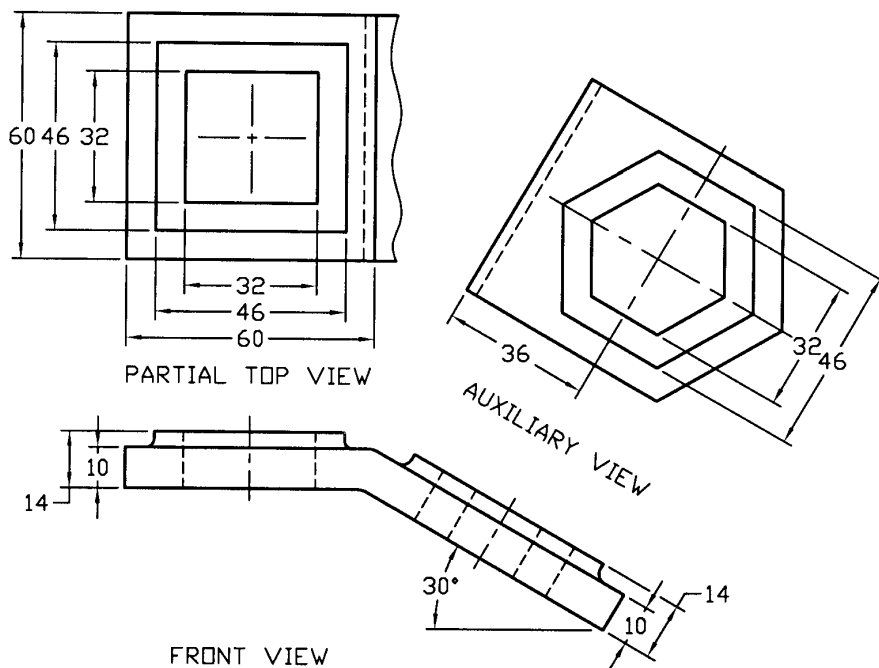
Solutions for Example problems

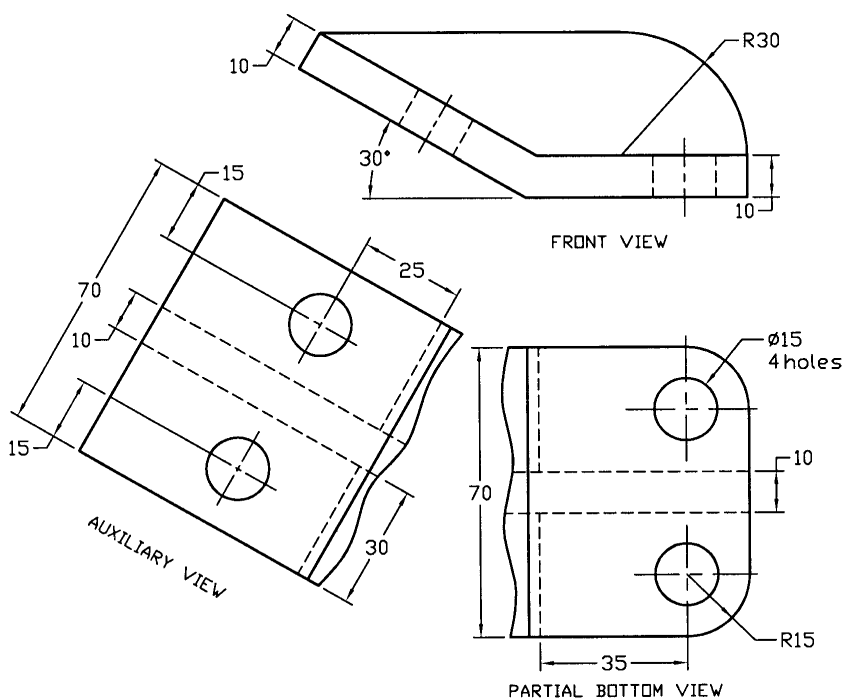
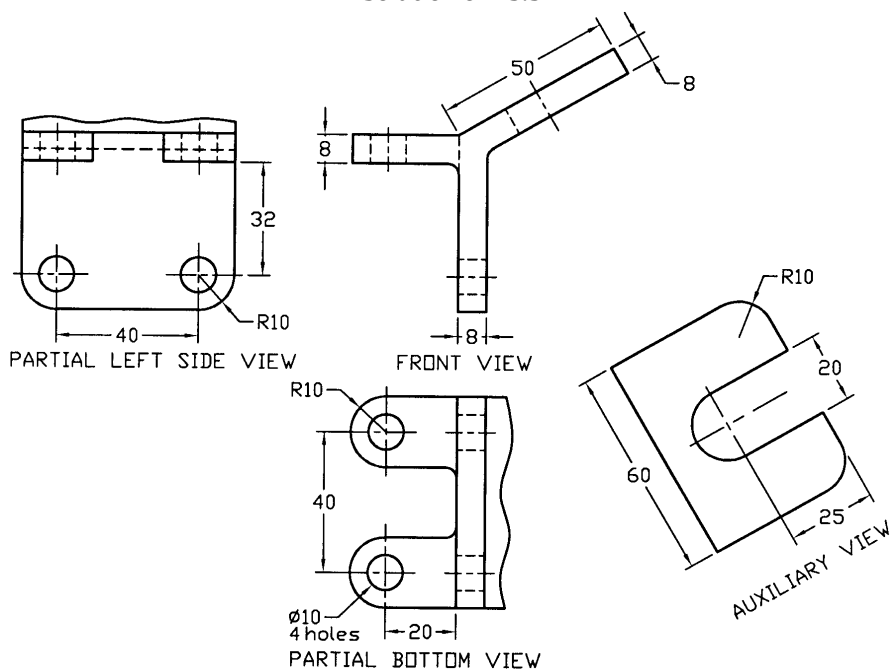


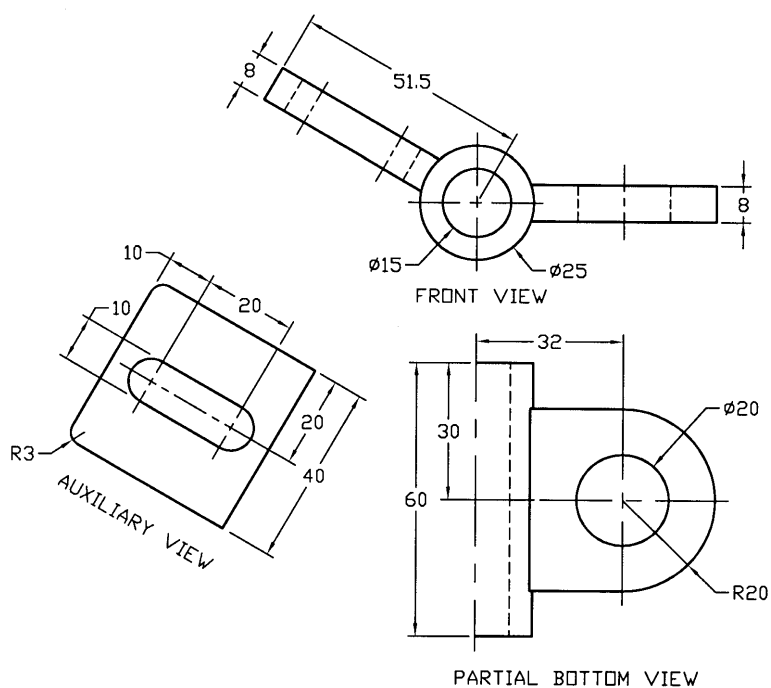
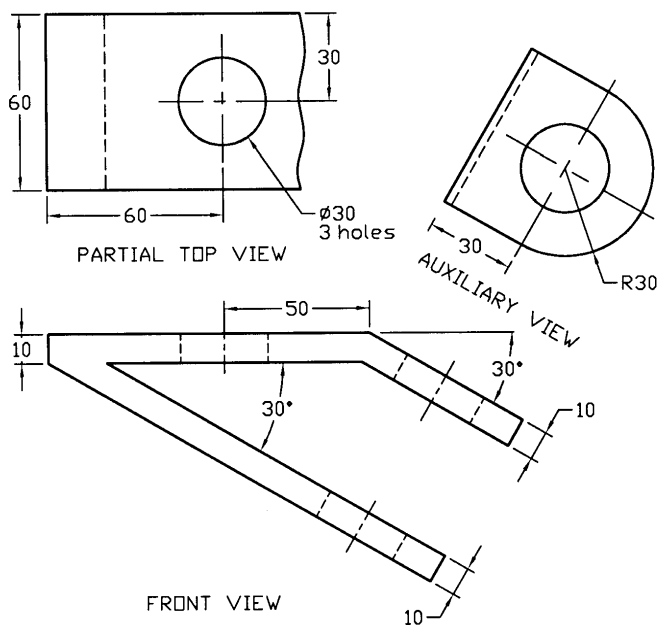
Solution of P5.1

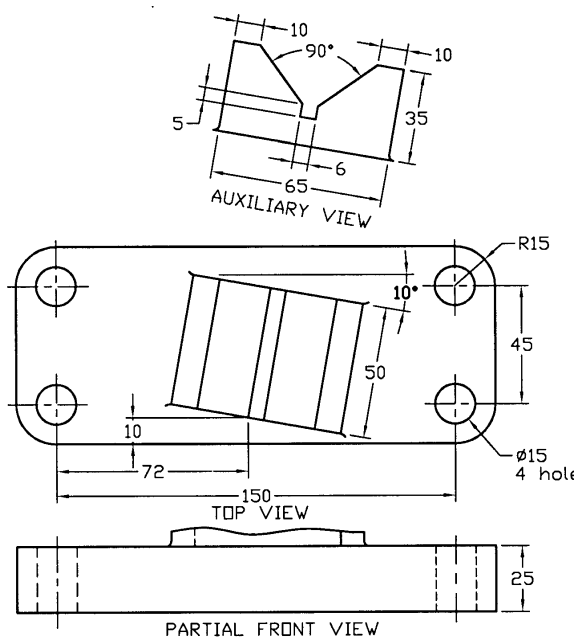
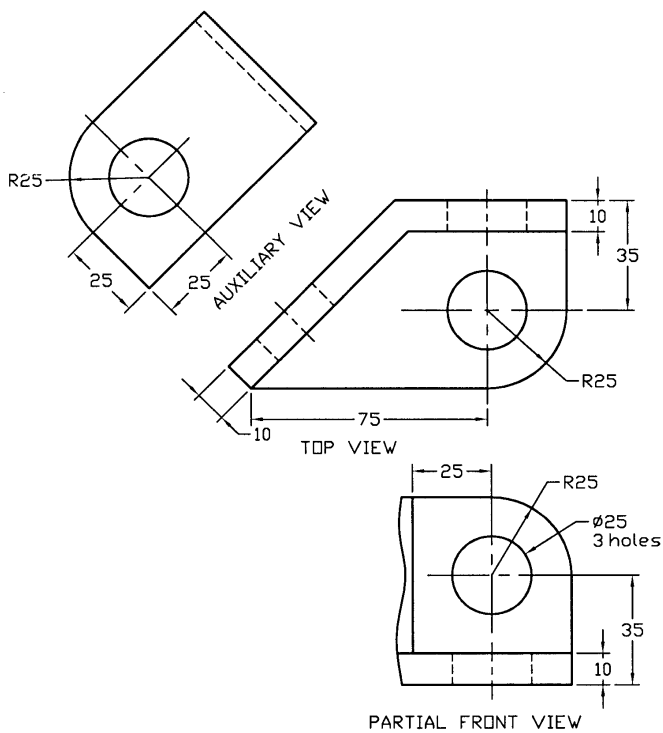


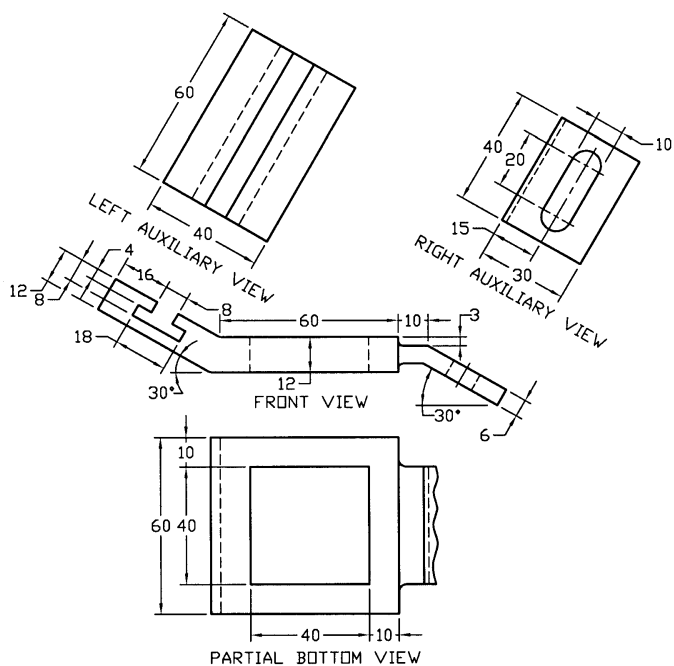
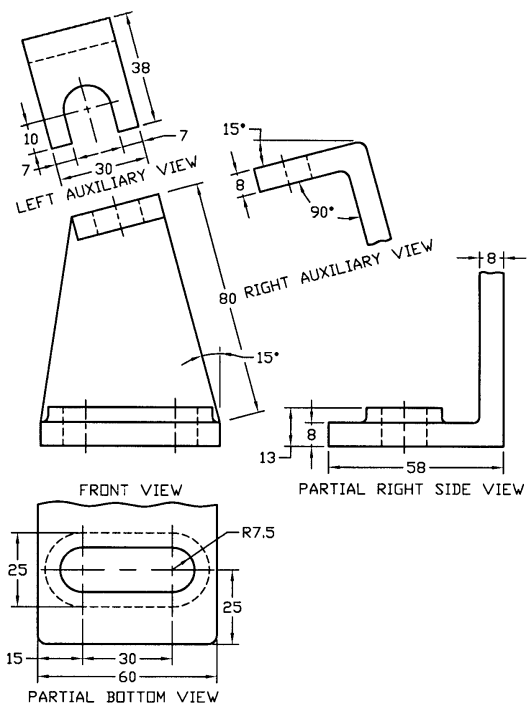
Solution of P5.2

**Solution of P5.3****Solution of P5.4**

**Solution of P5.5****Solution of P5.6**

**Solution of P5.7****Solution of P5.8**

**Solution of P5.9****Solution of P5.10**

**Solution of P5.11****Solution of P5.12**

Problems

Prob. 5.13: Draw front, partial top and auxiliary views of the guide block shown in Fig. P5.13.

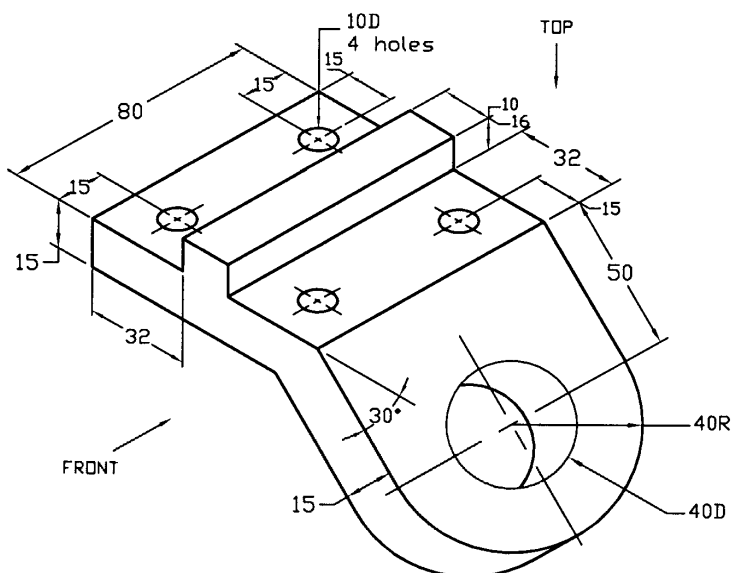


Fig. P5.13

Prob. 5.14: Draw front, partial top, partial right side and auxiliary views of the mounting bracket shown in Fig. P5.14.

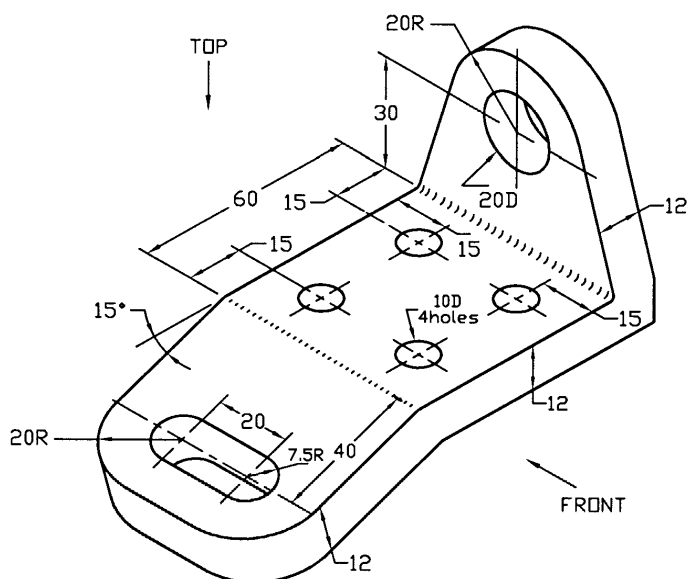


Fig. P5.14

Prob. 5.15: Draw front, partial top and auxiliary views of the clip shown in Fig. P5.15.

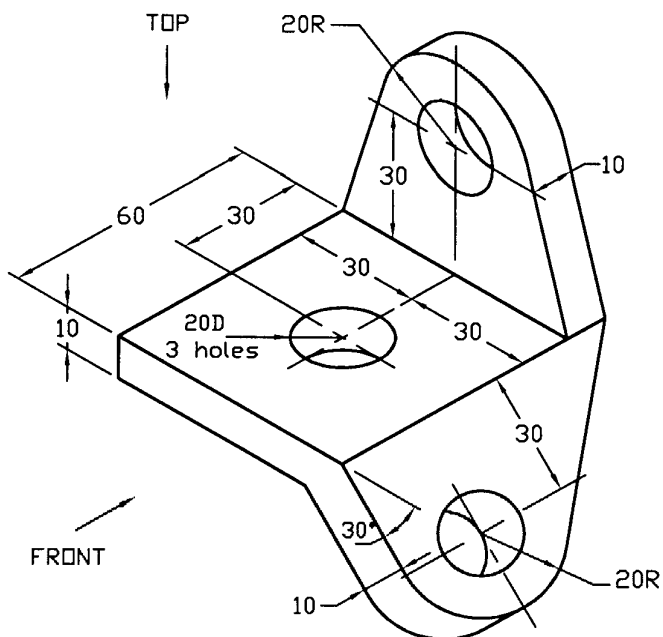


Fig. P5.15

Prob. 5.16: Draw the necessary views of the support base shown in Fig. P5.16.

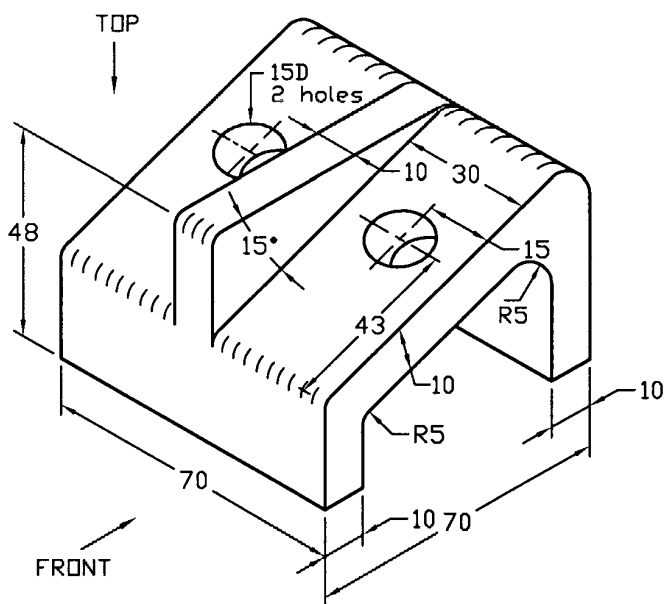


Fig. P5.16

Prob. 5.19: Draw the necessary views of the clip shown in Fig. P5.19.

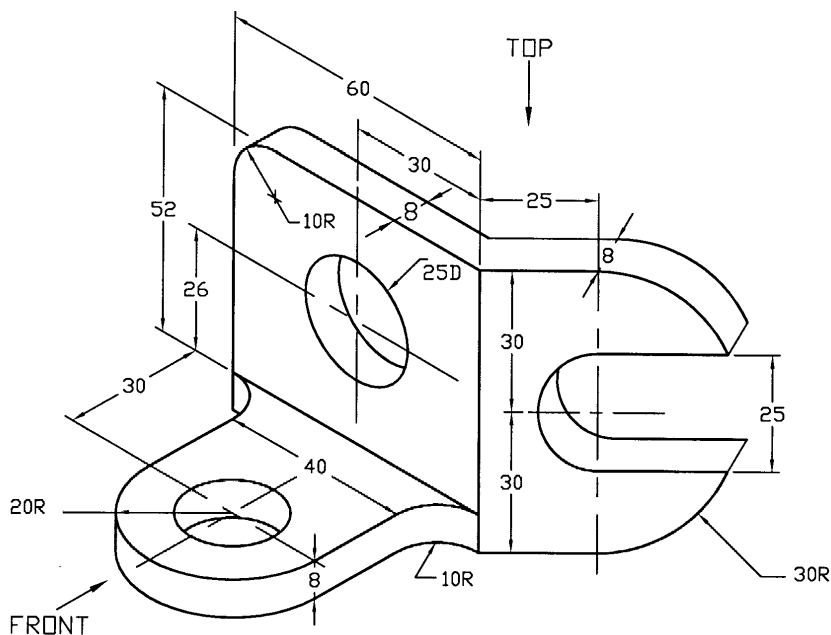


Fig. P5.19

Prob. 5.20: Draw the necessary views of the angle bracket shown in Fig. P5.20.

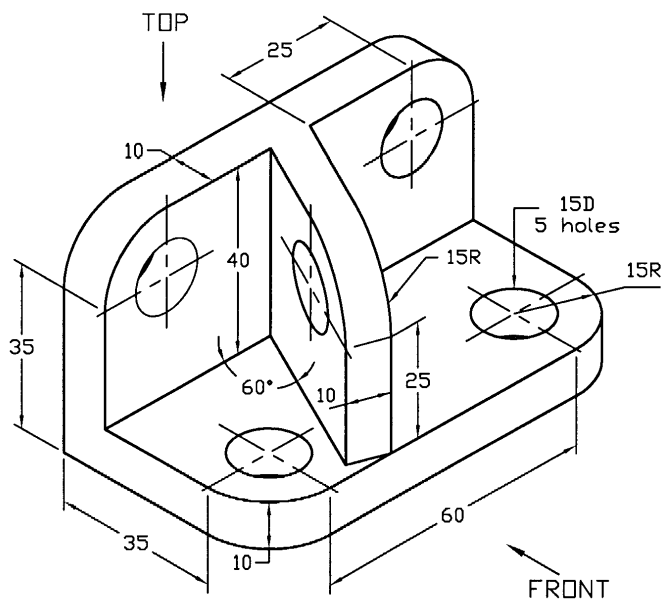


Fig. P5.20

Prob. 5.21: Draw the top, partial front and auxiliary views of the pin support base shown in Fig. P5.21.

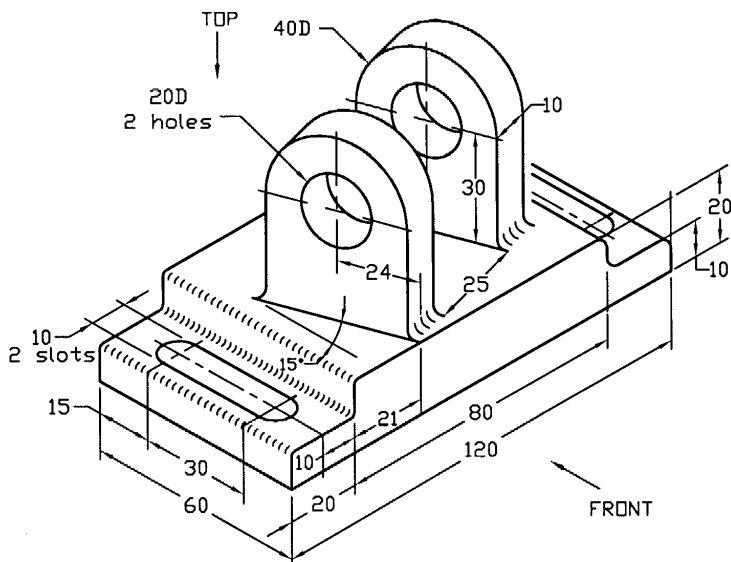


Fig. P5.21

Prob. 5.22: Draw the necessary views of the elbow shown in Fig. P5.22.

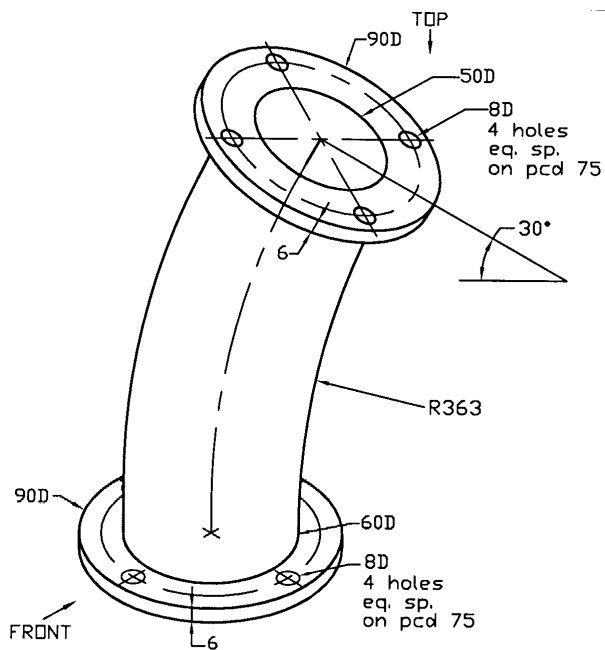


Fig. P5.22

Prob. 5.23: Draw the necessary views of the fork bracket shown in Fig. P5.23.

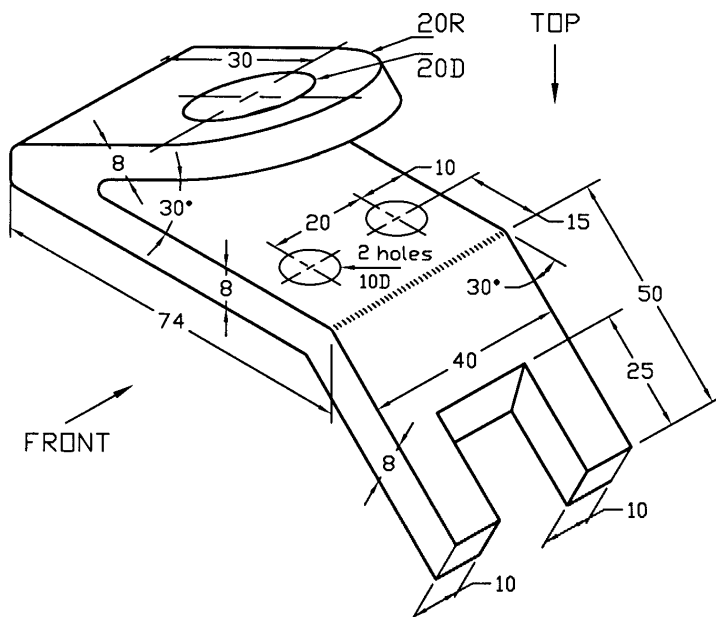


Fig. P5.23

Prob. 5.24: Draw the necessary views of the duct shown in Fig. P5.24.

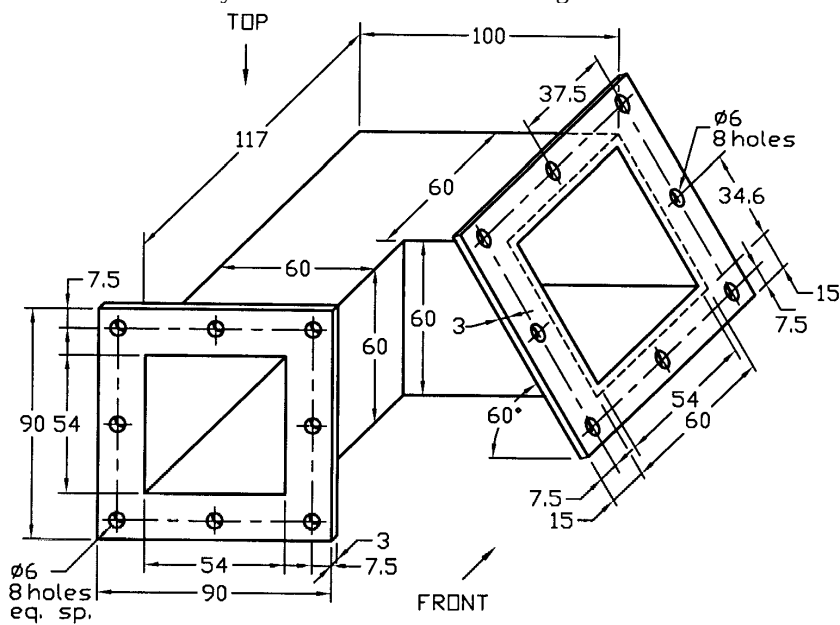


Fig. P5.24

Prob. 5.25: Draw the necessary views of the angle frame as shown in Fig. P5.25.

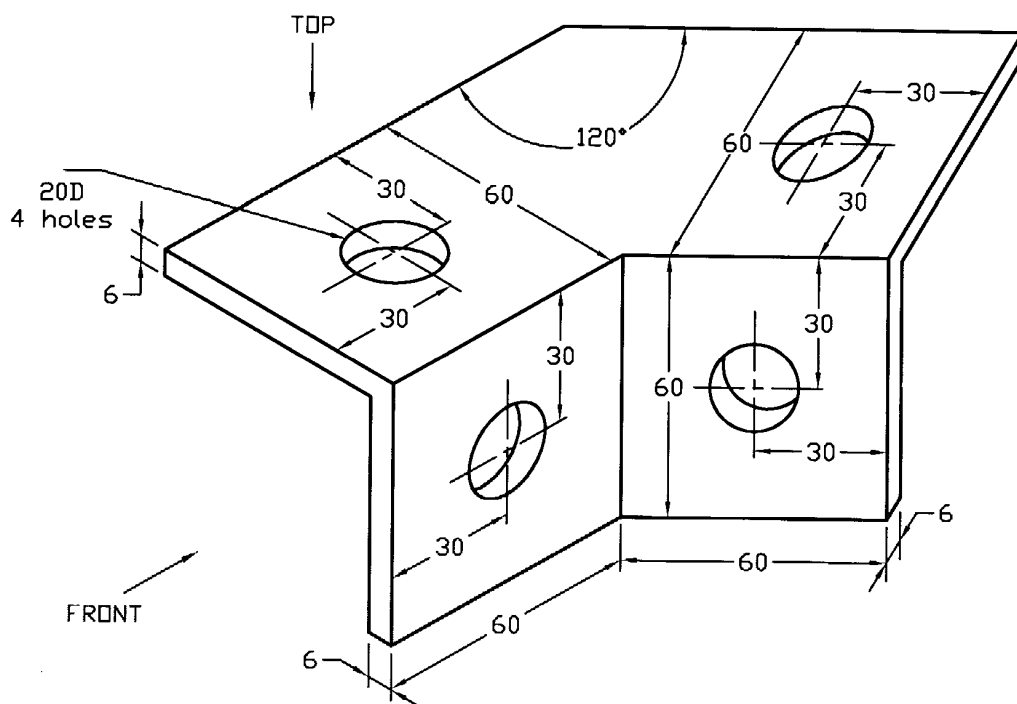


Fig. P5.25